Omar Alkhadra

Chicago, IL — (571) 278-9228 — omar.alkhadra@outlook.com — LinkedIn — Portfolio — U.S. Citizen

Education

University of Southern California — Los Angeles, CA
Master of Science in Applied Data Science (4.0/4.0) — Aug 2023 – May 2025

George Washington University — Washington, D.C.
Bachelor of Science in Systems Engineering, Minor in Statistics (3.64/4.0) — Aug 2019 – May 2023

Experience

GEICO - Pricing — Chicago, IL
Data Scientist — Jun 2025 – Present

- Developed XGBoost and logistic regression models predicting attrition probabilities under rate changes and policy characteristics; 3× reduction in combined calibration error (Brier, ECE, log loss) against the incumbent baseline
- Built stochastic simulation tool with interactive dashboard projecting policyholder attrition under rate-change scenarios, now used to inform pricing decisions across most state markets in production

University of Southern California — Los Angeles, CA
Machine Learning Researcher — Dec 2024 – May 2025

- Co-authored paper published at ACM SIGSPATIAL '25 on probabilistic assignment of noisy GPS trajectories to candidate locations under spatial uncertainty
- Reconstructed proprietary probabilistic baselines (SafeGraph's multi-stage KDE pipeline, Nishida's ILP formulation with spatiotemporal priors) as reproducible benchmarks

AXA XL — Los Angeles, CA
Cyber Intern – Data Science — Jun 2024 – Aug 2024

- Initiated a claim-prediction project with the actuarial team, building multiple ML classifiers on 10,000+ policies to model filing risk from underwriter-assigned categoricals and historical claims data
- Ran multivariate analysis and hypothesis testing to surface the features most associated with claim filing; identified key risk indicators and translated them into visualizations for underwriting and product teams

George Washington University — Washington, D.C.
Research Assistant — Aug 2022 – May 2023

- Co-authored peer-reviewed paper in *Environment Systems and Decisions*; built dynamic inoperability input-output model (Leontief) to quantify economic losses across interdependent sectors facing drought
- Conducted sensitivity analysis of policy intervention scenarios with expert-elicited parameters; validated model outputs against published economic loss estimates

Projects

Probabilistic Forecasting & Prediction Market Inefficiency Research

- Built conditional density model of numerical weather prediction errors: pooled XGBoost with Student-t residuals and ensemble-spread-conditional variance scaling; deployed as autonomous daily-retraining production system
- Validated calibration via PIT (KS=0.033) and Diebold-Mariano tests (p=0.006) against market-implied benchmarks; implemented robust decision rules from the literature (Lam 2016) to penalize overconfidence

Skills

- **Programming & ML:** Python (pandas, NumPy, SciPy, XGBoost, scikit-learn, PyTorch), SQL, R
- **Tools & Infrastructure:** Git, Linux, Jupyter, Snowflake, Apache Spark, REST APIs, AWS, Docker
- **Quantitative:** Probabilistic Calibration, Monte Carlo Methods, Stochastic Modeling, Time Series Analysis

Publications

- Saxena, N., Hsu, S., **Alkhadra, O.**, Shetty, M., Shahabi, C., & Horn, A. L. (2025). POIFormer: A Transformer-Based Framework for Accurate and Scalable Point-of-Interest Attribution. *SIGSPATIAL '25*, 607–619. ACM.
- Santos, J. R., **Alkhadra, O.**, Makrides, S., Huang, Y., & Menta, A. (2025). Drought impacts across economic sectors with policy analysis of mitigations. *Environment Systems and Decisions*.